

Automated Retinal Blood Vessel Segmentation Using Image Processing and U-Net

Full Project Plan — 22AIE437 Medical Image Processing

1. Project Idea (One-Line Summary)

Take raw retinal fundus images → enhance and clean them using classical image processing → segment the blood vessels pixel-by-pixel using a U-Net deep learning model → evaluate how accurately the model traces the vessel structure compared to expert-labeled ground truth.

The problem: Doctors look at retina scans (photos of the back of your eye) to check for diseases like diabetic retinopathy, glaucoma, etc. A big clue is the pattern of tiny blood vessels in the eye — how they branch, if they're leaking, if they're abnormal. But manually tracing every single blood vessel in a photo, by hand, is slow and tedious for doctors.

What we're building: A program that looks at a retina photo and automatically draws/highlights all the blood vessels for you — like an AI that traces them instead of a human doing it by hand.

Input → Output, simply:

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[Photo of retina, full color] → [Black & white image where  
                                WHITE = blood vessel  
                                BLACK = everything else]
```

How we get there — 2 phases:

Phase 1: Clean up the image (basic image processing) The raw photo is messy — reddish tones, uneven lighting, some noise. So before we do anything smart, we:

- Pull out just the "green" version of the image (turns out blood vessels show up clearest in the green channel, weirdly)
- Boost the contrast so faint vessels become more visible (a technique called CLAHE)
- Remove random noise/speckles

Think of this like adjusting brightness/contrast/sharpness on a photo before editing it further.

Phase 2: Let AI trace the vessels (deep learning) We feed the cleaned image into a model called **U-Net**. Think of U-Net like this: it first "zooms out" and looks at the whole image to

understand general shapes and context, then "zooms back in" to precisely mark, pixel by pixel, which parts are vessels and which aren't. It's like: get the big picture first, then get precise.

We train this model on a dataset called **DRIVE** — 40 retina photos where doctors have already manually traced the vessels by hand. The AI learns by comparing its guesses to the doctor's actual tracings, and slowly gets better.

The 2 review structure, explained simply:

- **Review 1:** "Build the basic version." We follow the original U-Net paper exactly, train it, and see how well it can trace vessels on its own. Simple version, straightforward implementation.
- **Review 2:** "Make it smarter." We upgrade the model by adding something called an "attention mechanism" — basically, we're teaching the AI to pay extra attention to the hard parts (like really thin, faint vessels it was missing before) instead of treating every pixel equally. Then we compare: did the upgraded version actually do better than the basic one?

End result you'll be able to show:

- Before/after image processing pictures
- The AI's vessel-tracing output next to the doctor's real tracing (so people can visually see how close it got)
- A score (like accuracy %) showing how good the AI is
- A comparison showing your "smarter" Review 2 version beat your "basic" Review 1 version

That's the whole project in a nutshell — clean the photo → teach an AI to trace vessels → make the AI better → prove it with numbers and pictures.

2. Dataset

Primary Dataset: DRIVE (Digital Retinal Images for Vessel Extraction)

- **Link:** <https://drive.grand-challenge.org/>
- **Contents:** 40 color fundus photographs (20 for training, 20 for testing), each 584×565 pixels
- Each image has a manually segmented vessel mask (ground truth) done by trained experts
- Also includes a field-of-view (FOV) mask for each image

Optional / Extension Datasets (use in Review 2 if you want multi-dataset validation)

- STARE: <https://cecas.clemson.edu/~ahoover/stare/>
- CHASE_DB1: <https://blogs.kingston.ac.uk/retinal/chasedb1/>
- HRF (High-Resolution Fundus): <https://www5.cs.fau.de/research/data/fundus-images/>

You do **not** need all of these. DRIVE alone is enough for both reviews. Add STARE/CHASE only if your sir wants broader validation.

3. Reference Papers

Review 1 Paper (Baseline — implement exactly)

"U-Net: Convolutional Networks for Biomedical Image Segmentation" Olaf Ronneberger, Philipp Fischer, Thomas Brox — MICCAI 2015

- Paper link: <https://arxiv.org/abs/1505.04597>
- PDF: <https://arxiv.org/pdf/1505.04597>

This is the foundational architecture paper. It's not retina-specific, but it's the base architecture every retinal vessel segmentation model builds on — this is your "correctly implement the reference paper" deliverable.

Domain-specific companion paper (cite alongside it to justify applying U-Net to retina):

Liskowski, P., & Krawiec, K. — *"Segmenting Retinal Blood Vessels with Deep Neural Networks,"* IEEE Transactions on Medical Imaging, 2016.

Review 2 Paper (Upgrade — pick ONE)

Option A — Spatial Attention U-Net (recommended) *"SA-UNet: Spatial Attention U-Net for Retinal Vessel Segmentation"*

- Paper (arXiv): <https://arxiv.org/pdf/2004.03696>
- Reports state-of-the-art results on DRIVE and CHASE_DB1

Option B — Compare multiple U-Net variants No single paper — instead frame Review 2 as a comparative study using:

- U-Net (your Review 1 baseline)
- Res-U-Net
- Attention U-Net

- Residual Attention U-Net Base architecture references: Zhang & Liu (2017) *Road Extraction by Deep Residual U-Net*; Oktay et al. (2018) *Attention U-Net*

Option C — Channel Attention Residual U-Net "*Channel Attention Residual U-Net for Retinal Vessel Segmentation*"

- Paper (arXiv): <https://arxiv.org/pdf/2004.03702>
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4. GitHub Repositories

For Review 1 (plain U-Net baseline on DRIVE)

1. https://github.com/CVxTz/medical_image_segmentation Implements a U-Net for eye vessel segmentation on DRIVE, reporting AUC-ROC of 0.982. Good, clean, minimal starting point.
2. <https://github.com/onurboyar/Retinal-Vessel-Segmentation> U-Net based segmentation using DRIVE and STARE, with a focus on data augmentation strategy — useful if you want to show augmentation experiments.
3. <https://github.com/33-Papers/U-Net> A from-scratch PyTorch replication of the original U-Net paper — good if your sir wants you to show you understand the architecture itself, not just use someone else's pipeline.

For Review 2 (upgraded model)

4. <https://github.com/clguo/SA-UNet> Official code for Spatial Attention U-Net — matches Option A above. Trains/evaluates on DRIVE and CHASE_DB1.
5. <https://github.com/arkanivasarkar/Retinal-Vessel-Segmentation-using-variants-of-UNET> Implements U-Net, Res-U-Net, Attention U-Net, and Residual Attention U-Net together, tested on DRIVE, STARE, and HRF — matches Option B above (comparative study).

Important: Don't just clone and run these. Read the code, understand each block (data loader, model architecture, training loop, loss function, evaluation metrics), and re-implement/modify key parts yourself so you can explain every line in your viva.

5. Full Project Workflow (Start to Finish)

STEP 1: Data Collection

Download DRIVE dataset (40 fundus images + vessel masks + FOV masks)

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STEP 2: Preprocessing (classical image processing – Unit 2)

- a) Extract Green channel (vessels show highest contrast in green channel vs red/blue – red is too bright, blue is too noisy)
- b) Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) to enhance local contrast without amplifying noise
- c) Apply Gaussian/Median filtering for noise removal
- d) Normalize pixel values (0-1 range)
- e) Apply Gamma correction (optional, improves visibility of thin vessels)

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STEP 3: Data Preparation for Deep Learning

- a) Patch extraction (since DRIVE has only 20 training images, extract random small patches, e.g., 48x48 or 64x64, to create thousands of training samples)
- b) Data augmentation: rotation, flipping, elastic deformation
- c) Train/validation split

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STEP 4: Model Building – U-Net (Review 1)

Encoder (contracting path): Conv → ReLU → Conv → ReLU → MaxPool
(repeated, doubling filters each level: 64 → 128 → 256 → 512 → 1024)

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Bottleneck

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Decoder (expanding path): UpConv → Concatenate with skip connection
→ Conv → ReLU → Conv → ReLU (repeated, halving filters each level)

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Final 1x1 Conv → Sigmoid → Binary vessel mask output

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STEP 5: Training

Loss function: Binary Cross-Entropy or Dice Loss (or a combination)
Optimizer: Adam (learning rate ~0.001)
Epochs: 50–150 depending on compute available

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STEP 6: Evaluation (Review 1)

Metrics: Accuracy, Sensitivity (Recall), Specificity, F1/Dice score,
AUC-ROC – compare against DRIVE's expert ground truth masks

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===== END OF REVIEW 1 =====

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STEP 7: Upgrade the Model (Review 2)

Replace/extend plain U-Net with your chosen upgrade:

- Add spatial attention gates at skip connections (SA-UNet), OR
- Add residual blocks + attention gates (Attention/Res U-Net), OR
- Add channel attention mechanism (CAR-UNet)

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STEP 8: Re-train and Re-evaluate

Same metrics as Step 6, now compare:

Review 1 model (plain U-Net) vs Review 2 model (upgraded)

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STEP 9: Result Comparison and Analysis

Show side-by-side segmented vessel masks (visual comparison)

Show metrics table: which model performed better and why

Discuss where the upgraded model improved (e.g., thin/capillary vessels, reduced false positives near optic disc)

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STEP 10: (Optional) Build a simple demo

Streamlit/Gradio app where you upload a fundus image and it outputs the segmented vessel mask – great for viva demonstration

6. Review-wise Plan

📌 Review 1 Plan — "Implement the reference paper exactly"

Task	Details
Reference paper	Ronneberger et al., U-Net (MICCAI 2015) — https://arxiv.org/abs/1505.04597
Dataset	DRIVE — https://drive.grand-challenge.org/
GitHub base	CVxTz/medical_image_segmentation or 33-Papers/U-Net
What to build	Preprocessing pipeline (green channel + CLAHE + noise removal) + plain U-Net architecture trained on DRIVE
What to present	Architecture diagram, preprocessing steps with before/after images, training curves (loss/accuracy vs epochs), final metrics (Accuracy, Sensitivity, Specificity, AUC, Dice)
What to say to your sir	"We reproduced the original U-Net architecture from Ronneberger et al. (2015) and applied it to retinal vessel segmentation on the DRIVE dataset, following preprocessing techniques consistent with Liskowski & Krawiec (2016)."

📌 Review 2 Plan — "Upgrade on top of Review 1"

Task	Details
Reference paper	SA-UNet (or your chosen Option B/C) — https://arxiv.org/pdf/2004.03696

Task	Details
What changes	Add spatial attention module at skip connections (or switch to Res/Attention U-Net comparison)
What to build	Modified U-Net architecture + same training/eval pipeline from Review 1
What to present	Comparison table: Review 1 (plain U-Net) vs Review 2 (upgraded) on the same metrics; visual comparison of predicted masks; explanation of WHY the upgrade helps (attention mechanism focuses on thin/low-contrast vessel regions that plain U-Net misses)
What to say to your sir	"Building on our Review 1 U-Net baseline, we implemented spatial attention gates as proposed by Guo & Bertani (SA-UNet, 2020), improving vessel segmentation accuracy particularly on thin capillaries, achieving a Y% improvement in Dice score over our baseline."

7. Tools You'll Need

- **Language:** Python
 - **Frameworks:** Keras/TensorFlow (SA-UNet repo uses Keras 2.3.1 + TensorFlow 1.14) or PyTorch (most other repos use PyTorch — recommended since more actively maintained)
 - **Libraries:** OpenCV (for CLAHE, filtering), NumPy, scikit-image, matplotlib, scikit-learn (for metrics)
 - **Compute:** Google Colab (free GPU) is sufficient for DRIVE since it's a small dataset (only 20 training images + patches)
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8. Quick Checklist Before Review 1

- ☐ DRIVE dataset downloaded and organized (train/test/masks/FOV folders)
- ☐ Green channel extraction + CLAHE + noise removal implemented and visualized
- ☐ Patch extraction / augmentation pipeline working
- ☐ Plain U-Net model built and trained
- ☐ Evaluation metrics computed and compared to paper's reported numbers
- ☐ Report written citing Ronneberger et al. (2015) and Liskowski & Krawiec (2016)

9. Quick Checklist Before Review 2

- ☐ Attention/residual module added to U-Net architecture
- ☐ Retrained on same DRIVE split as Review 1 (fair comparison)

- ☐ Metrics table comparing Review 1 vs Review 2 results
- ☐ Visual mask comparison images prepared
- ☐ Report written citing your chosen upgrade paper (SA-UNet / CAR-UNet / Attention U-Net)
- ☐ (Optional) Simple demo app (Streamlit/Gradio) ready for live viva demonstration